Name: ______________________

SURDS,
(MULTIPLYING AND DIVIDING SURDS)

Date: ________________

LO: I can multiply and divide surds, simplifying where possible.

Multiplying and dividing surds

To **multiply** surds: a × b = (ab). To **divide**: a ÷ b = (a/b). Always **simplify** the result.

Key fact: a × a = a. This is useful for rationalising and expanding brackets.

Quick examples

	3 × 5 = 15	multiply radicands
	6 × 6 = 6	same surd squared
	20 ÷ 5 = 4 = 2	divide radicands
	3 × 5 = 8	multiply, not add: 15
	2 3 × 4 3 = 8 9	= 8 × 3 = 24

Key idea

Multiply: a × b = (ab). Divide: a ÷ b = (a/b). Simplify the result.

a × b = (ab) | a ÷ b = (a/b)

Always simplify the final answer

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – multiplying surds

Simplify 3 × 12.

3 × 12 = (3 × 12)

= 36

= 6

6

Multiply the numbers under the root, then simplify.

Example 2 – with coefficients

Simplify 2 5 × 3 2.

Multiply coefficients: 2 × 3 = 6

Multiply surds: 5 × 2 = 10

= 6 10

6 10

Multiply the numbers outside together and the surds together.

Example 3 – dividing surds

Simplify 72 ÷ 2.

72 ÷ 2 = (72/2)

= 36

= 6

6

Divide the radicands, then simplify.

Example 4 – expanding brackets

Expand and simplify 3(2 + 3).

3 × 2 = 2 3

3 × 3 = 3

= 2 3 + 3

2 3 + 3

Distribute the surd across the bracket.

YOUR TURN – PRACTICE (deliberate practice)

1. Multiplying simple surds

Simplify each product.

a) 2 × 8

b) 5 × 5

c) 3 × 7

d) 6 × 24

2. With coefficients

Simplify each product.

a) 3 2 × 4 2

b) 2 3 × 5 3

c) 4 5 × 2 3

d) 3 7 × 7

3. Dividing surds

Simplify each division.

a) 50 ÷ 2

b) 48 ÷ 3

c) 6 10 ÷ 2 5

d) 75 ÷ 3

4. Expanding brackets

Expand and simplify.

a) 2(3 + 2)

b) 5(4 - 5)

c) 2 3(1 + 3)

d) 7(7 + 2 3)

5. Mixed multiplication and division

Simplify each expression.

a) 2 × 3 × 6

b) $(2\ 3)^2$

c) 45 ÷ 5

d) (5 + 1)(5 - 1)

e) (3 + 2)(3 - 2)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) a × b = (ab) ☐

Correct answer: __________

Reason: __________________

b) 3 × 5 = 8 ☐

Correct answer: __________

Reason: __________________

c) a × a = a ☐

Correct answer: __________

Reason: __________________

d) $(2\ 3)^2 = 2\ 9 = 6$ ☐

Correct answer: __________

Reason: __________________

e) a ÷ b = (a/b) ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Expand and simplify **(3 + 5)(3 - 5)**. What do you notice about the result? Use this to explain why $(a + b)(a - b) = a^2 - b^2$.

Expression: ____________________________

Simplified: ____________________________

b) Show that **$(2 + 3)^2 = 5 + 2\ 6$**. Hence find the value of **$(2 + 3)^2 + (2 - 3)^2$**.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!